

5.1: The FPC's approach to assessing risks from market-based finance and building resilience

MBF plays a key role in the provision of credit to the real economy, so it is important that vulnerabilities in it are identified and addressed.

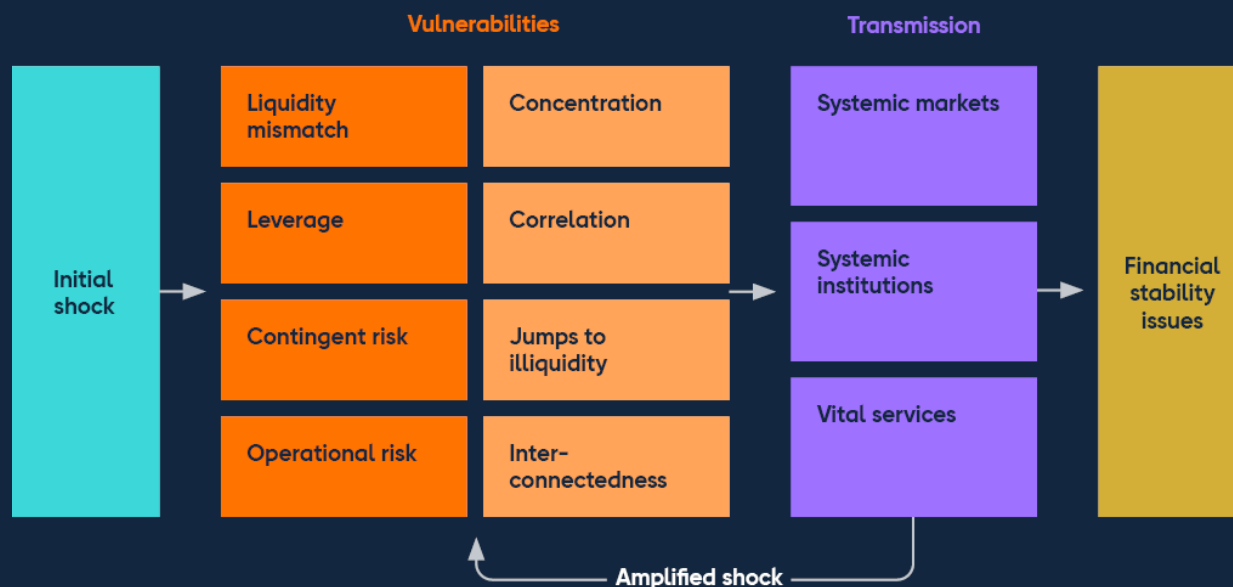
MBF is the system of markets, NBFIs and infrastructure which, alongside banks, provides financial services to support the wider economy. NBFIs such as investment funds and insurers are connected to each other and other parts of the financial system, including banks. They have come to play an increasingly central role in the UK financial system, now accounting for around half of total financial sector assets and responsible for more than half of the lending to UK corporates. This has diversified the supply of finance for UK businesses. But it also makes it important that MBF is resilient enough to absorb, and not amplify, financial and economic shocks, so that it can continue to support the provision of financial services to UK households and businesses. Policy measures to improve resilience can help reduce the likelihood that vulnerabilities in MBF cause wider disruption, and also reduce the impact of such disruption if it occurs.

The FPC recently set out its approach to identifying and assessing risks associated with MBF, building resilience, and responding when disruption occurs.

In particular, it detailed the types of vulnerabilities that the FPC considers (see Figure 5.1). 'Microfinancial' vulnerabilities relate to particular NBFI business models that can make individual firms vulnerable to shocks. For example, liquidity mismatch – where a firm's assets are less liquid than its liabilities – is a type of vulnerability inherent in some business models, which can amplify shocks to the demand for liquidity. This type of vulnerability can interact with 'macrofinancial' vulnerabilities, which relate to market structure and the collective behaviour of firms. An example of this type of vulnerability is correlated positions, which can mean a whole sector has similar strategies and therefore acts in the same way during stress, thus amplifying it. Vulnerabilities can be of concern from a financial stability perspective if they have the capacity to adversely impact systemic markets, [12] institutions or the provision of vital services.

Figure 5.1: How vulnerabilities in MBF can affect financial stability

Vulnerabilities in MBF and transmission channels to financial stability



5.2: Developments in vulnerabilities in market-based finance

Using the analytical framework above, the FPC has previously identified several vulnerabilities in the system of MBF, which remain significant. These include:

- liquidity mismatch in MMFs and open-ended funds (OEFs);
- leverage in NBFIs, for example in LDI funds and the US Treasury cash-futures basis trade;
- liquidity demands from margin calls in times of stress; and
- the capacity of markets to intermediate in stress, so-called jump-to-illiquidity risk.

These vulnerabilities could crystallise in the context of the current environment of elevated and volatile interest rates, amplifying any tightening in financial conditions. That said, MBF has so far been broadly resilient to recent market volatility and higher interest rates.

OEFs for riskier corporate credit, such as high-yield bonds and leveraged loans, have seen investor outflows over the past two years. These have so far been orderly.

Liquidity mismatches can arise when assets are less liquid than longer-dated liabilities, as is the case for many OEFs. This liquidity mismatch creates a risk of investor outflows overwhelming their capacity to liquidate assets in an orderly manner, without forced selling.

OEFs for riskier corporate credit, such as high-yield bonds and leveraged loans, have seen investor outflows over the past two years. For example, sterling high-yield bond funds have seen outflows of around 10% of assets under management since January 2022. These outflows have so far been orderly. See Box B for further discussion of high-yield bonds and leveraged loans.

Open-ended property funds invest in illiquid commercial real estate (CRE) assets while offering much shorter-term redemptions. They remain challenged by the headwinds facing the CRE market, and some have suspended redemptions due to the level of outflows. Suspensions or redemption deferrals are used by funds to limit the risk of outflows triggering forced selling, but these tools can have the potential adverse effect of worsening market sentiment, hence triggering further redemptions or sales. So far these developments have not spilled over to the wider CRE market.

As set out in the July FSR, OEFs comprise only a small proportion of the total investor base in UK CRE, which limits the market impact that their asset disposals might have.

A sharp investor reappraisal of credit risk among highly leveraged corporates could crystallise vulnerabilities in the system of MBF and have implications for the real economy.

The opacity and the lack of frequent re-pricing of private credit assets increases their vulnerability to sharp and correlated falls in value. And given the common features of private credit and leveraged loans – such as the floating rate nature of the lending and links to private equity activity – there is a risk of correlated stresses in these and other interconnected markets, such as private equity. Fire selling and reductions in risk appetite could be amplified by NBFIs with liquidity mismatches and leveraged positions. See Box B for further discussion of this topic.

Hedge fund net short positioning in US Treasury futures has recently increased, as asset manager net long positions have continued to grow...

As an alternative to investing directly in US Treasury bonds, asset managers can gain similar exposures via futures contracts linked to the underlying bonds. The synthetic leverage provided by futures requires less initial cash outlay, freeing up resources for other purposes, such as meeting redemptions or investing in higher yielding assets. This demand for futures pushes up their prices relative to the prices of corresponding cash bonds. The ‘cash-futures basis trade’ exploits these price differences. In normal market conditions, the basis trade plays a useful role by intermediating between different types of market participants, helping to keep the cash and futures markets broadly aligned, thus improving market liquidity and efficiency.

This intermediation can, however, amplify market stress. To boost returns, hedge funds tend to use significant amounts of leverage, supported by short-term borrowing in the repo market. And relatively low initial margin requirements for the futures contracts enable potentially high degrees of leverage. Sharp increases in volatility in market interest rates could lead to increases in margin required on the futures positions, or hedge funds may find it harder to refinance their borrowing in the repo market. This, combined with any breaches of risk- or loss-limits, could force funds rapidly to unwind their positions (as seen in the March 2020 ‘dash for cash’ episode), which could amplify the market volatility. Since the July FSR, hedge funds’ net short positioning in US Treasury futures has increased further, from around US\$650 billion to around US\$800 billion (Chart 5.1), now exceeding in nominal terms its size prior to the 2020 market stress. This trend has, as expected, largely mirrored asset managers’ increased net long positioning.

Chart 5.1: Hedge fund net short positioning in US Treasury futures has recently increased further, as asset manager net long positions have continued to grow

Aggregate net positioning across US Treasury futures



Sources: CFTC data and Bank calculations.

...and although the recent volatility in rates has not crystallised financial stability risks, vulnerabilities associated with it remain.

Structural changes could have mitigated some of the vulnerabilities associated with the trade since 2020. Higher initial margin requirements on Treasury futures, in response to increased market volatility, have had the effect of reducing the effective (synthetic) leverage available to market participants, since more cash is required upfront for a given level of exposure. For example, the [BIS](#) estimates that allowed leverage, defined as the contract price divided by initial margin, in five-year Treasury futures remains elevated but has fallen from around 175x prior to the pandemic to around 70x. Nonetheless it remains an important vulnerability which continues to be monitored by the FPC. There are potential spillover risks to the UK. These could be via the participation of the same hedge funds in both US Treasury and gilt markets, and also more indirectly via disruption to the systemically important US Treasury market.

The FPC routinely monitors hedge fund activity in core UK markets,[13] although there is no direct read-across of this trade to the UK gilt market, due to structural differences. For example, asset managers are more likely to use repo borrowing and swaps directly to gain leveraged exposure to interest rates.

5.3: Improving the resilience of market-based finance

Given the potential for vulnerabilities in MBF to pose risks to UK financial stability, the FPC seeks to build resilience in MBF in order to address them. The high degree of interconnectedness in MBF, including across borders, means that risks are most effectively addressed through internationally co-ordinated reforms. The FPC also works to reduce vulnerabilities domestically where this is effective and practical. See Table 5.A for an overview of progress in building resilience against key vulnerabilities in MBF.

Liquidity mismatch in MMFs has been identified as an important vulnerability and is a continued area of focus for the FPC.

The FPC judged [in March 2023](#) that MMFs holding more liquid assets would reduce financial stability risks associated with rapid investor redemptions. And Bank staff analysis suggests that around 50%–60% of total assets falling within the shorter-maturity ‘weekly liquid assets’ category would give sterling-denominated MMFs a high level of resilience to severe but plausible stress events. This analysis is based on the levels of investor redemptions seen during both the 2020 dash for cash and the 2022 gilt market stress.

The FPC welcomes the [proposals currently under consultation](#) to increase the resilience of UK-based MMFs, including increasing daily and weekly liquidity requirements to 15% and 50% respectively.

Many MMFs operate cross-border, and indeed around 90% of total sterling MMF assets are domiciled in the EU and hence are not subject to UK regulation and Financial Conduct Authority (FCA) supervision. It is therefore important that domestic policy work is complemented by work across jurisdictions, and in particular that relevant measures set out by the FSB to enhance MMF resilience are implemented by member jurisdictions. This will help to ensure an adequate level of resilience and mitigate the potential risks from cross-border spillovers and

regulatory arbitrage. The forthcoming stocktake by the FSB should help highlight progress made to date and areas where further work is required, ahead of a fuller assessment in 2026.

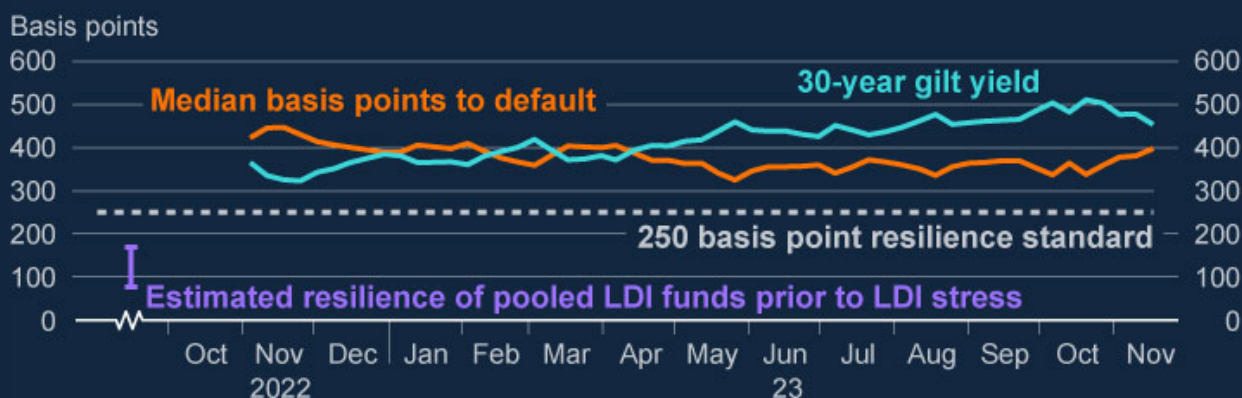
Against a backdrop of rising long-term interest rates, LDI funds have maintained their resilience to further rates shocks in accordance with FPC Recommendations.

LDI is an investment approach used by defined-benefit (DB) pension schemes to help ensure that the value of their assets (ie their investments) moves more in line with the value of their liabilities (ie the pensions they have promised to pay in the future). Following the gilt market stress in autumn 2022, in March 2023 the FPC judged that LDI funds should be resilient to a shock to the gilt yield curve of 250 basis points, at a minimum. This is additional to maintaining sufficient resilience to manage other risks and day-to-day movements in yields.

This resilience framework for LDI funds has been functioning broadly as intended in an environment of higher market interest rates. Long-dated gilt prices fell significantly over the first half of 2023, pushing up yields, and they have continued to display volatility since the July FSR (Section 1). Other things equal, falling gilt prices can reduce the resilience of LDI funds to further shocks as they result in margin and collateral calls and reduce the net asset value of the fund. Meeting these calls and rebalancing their portfolios may require additional capital to be injected by individual or groups of investor pension schemes. LDI funds in aggregate have maintained their resilience above the 250 basis point resilience standard by recapitalising in response to falling gilt prices (Chart 5.2). While resilience levels vary across funds, on average levered fixed and real pooled funds are resilient to an almost 400 basis point increase in yields.

Chart 5.2: While long-term gilt yields have risen, LDI funds have maintained sufficient resilience to shocks

Median basis points move in gilt yields that would cause leveraged UK pooled LDI funds to default and yields on 30-year gilts (a)



Sources: Bloomberg Finance L.P., FCA supervisory data and Bank calculations.

(a) The 250 basis point minimum resilience refers to the FPC's Recommendation in March 2023 that that LDI funds should be resilient to stresses that account for both historical volatility in gilt yields, and the potential for their forced sales to amplify market stress and disrupt gilt market functioning. The FPC judged that these factors imply that LDI funds should be resilient to a yield shock of around 250 basis points, at a minimum, in addition to the resilience required to manage other risks and day-to-day movements in yields. Estimated resilience is based on supervisory data.

While LDI funds are resilient in aggregate, there remain some areas for improvement for some funds to implement the relevant guidance fully. Some LDI managers' recapitalisation in response to falling gilt prices has been slower than the expected five days. Funds must also ensure that their management buffers and triggers take into account the length of time they need to recapitalise. It is therefore important that the work to put in place a monitoring and enforcement framework for pooled funds by the FCA and relevant international regulators continues.

While rising interest rates can require DB pension schemes to recapitalise LDI funds they may be invested in, as set out above, they nonetheless have the effect of improving the financial position of the pension schemes. This is because higher rates reduce the present value of a scheme's future liabilities (ie payments to

pensioners) relative to its assets. As a result of this dynamic, the funding position of DB pension schemes has improved considerably since interest rates began to rise, from being in deficit to being in surplus in aggregate.^[14]

The forthcoming SWES exercise will aid the FPC's understanding of how market-based finance behaves as an interconnected system.

An important aspect of the FPC implementing the framework in Figure 5.1 is developing a system-wide view that covers the interconnections between parts of the system of MBF, links from MBF to other parts of the financial system, and the response of the system as a whole to shocks (see [Kroszner \(2023\)](#)). The SWES exercise will provide the FPC with new information about the behaviours of non-bank financial institutions and banks during stressed financial market conditions, and how those behaviours might interact to amplify shocks to UK financial markets that are core to UK financial stability. It will do so by testing the ability of market participants to anticipate liquidity demands in stresses, their strategies for meeting those demands, the impact of leverage in amplifying liquidity imbalances during periods of market stress, and whether participating sectors' expectations of one another in periods of stress would be met. See Box E for more discussion of the recently published scenario underpinning the SWES exercise.

The Bank is working to expand its toolkit to backstop market functioning in core sterling markets to support financial stability in exceptional circumstances.

The priority of the FPC and authorities internationally is for individual market participants, and their regulators, to maintain appropriate resilience standards that allow the system of MBF to absorb and not amplify severe but plausible shocks that could threaten financial stability. But it is not feasible for MBF to maintain a level of resilience that would insure against all of the most extreme system-wide liquidity stresses. In such circumstances of system-wide stress, where reasonable levels of resilience in the market are likely to be exhausted, central bank tools can support financial stability by providing backstop liquidity.

The Bank is working to develop the tools to enable it to intervene where liquidity-related dysfunction in core sterling markets threatens financial stability (see [Ramsden \(2023\)](#) and [Hauser \(2023\)](#)). In system-wide stress scenarios where NBFIs are seeking temporary liquidity, it is preferable to backstop market

functioning by lending directly to NBFIs against high quality collateral, rather than with asset purchases. This presents less risk to public funds and to the stance of monetary policy, and less potential moral hazard. However, in some stress scenarios, such as where system-wide stress is driven by NBFIs' need to reduce exposures or deleverage, it may instead be appropriate to use temporary and targeted asset purchases. For example, during the Autumn 2022 stress in long-dated gilt markets, which involved fire selling by some NBFIs, no parties were able or willing to borrow from the Bank at sufficient speed to stop the fire-sale dynamic. On that basis, the Bank launched a targeted programme of asset purchases, which was fully unwound by January 2023 after the stress had abated.

The FPC considers that resilience standards for MBF should be developed in co-ordination with work to enhance central bank tools to respond in stress. This should ensure that public backstops do not substitute for a failure to achieve the appropriate level of private insurance.

Overall, the underlying vulnerabilities in the system of MBF, identified by the FPC and other financial stability authorities internationally, remain largely unaddressed. Absent actions to mitigate them, they could amplify any future shocks.

Table 5.A summarises the current status of policy initiatives to address identified key vulnerabilities in MBF domestically and internationally. The FPC continues strongly to support the FSB's international work programme to increase the resilience of MBF.

Table 5.A: Overview of progress on building resilience against key vulnerabilities in MBF

Vulnerability	Financial stability implications	Policy recommendations and next steps
Maturity mismatch arises when assets are less liquid or longer dated than liabilities.		
Money market funds (MMFs)	MMFs are used by UK corporates, investment funds, and other NBFIs as a way of managing cash balances. Investors hold £260 billion in sterling-denominated MMFs. Liquidity mismatch between the redemption terms and the liquidity of some of their assets makes MMFs vulnerable to sharp redemptions from investors in stress and so risks both runs and contagion across the sector. This could amplify shocks, impact financial stability if investors cannot access cash, and lead to tighter financial conditions for the economy.	In order to be sufficiently resilient, sterling MMFs should be able to withstand severe but plausible levels of outflows, without triggering runs or contagion that would create risks to financial stability. The FSB has agreed an international approach to MMFs, suggesting a range of measures such as higher liquid asset requirements or liquidity management tools to be implemented by national authorities globally. The FSB is due to publish a stocktake of progress in December 2023. UK authorities have launched a consultation paper on enhancing MMF resilience measures, including increasing daily and weekly liquidity requirements to 15% and 50% respectively.

Vulnerability	Financial stability implications	Policy recommendations and next steps
Open-ended funds (OEFs)	Globally, the assets under management of OEFs primarily investing in UK equities, sterling government bonds and sterling corporate bonds and UK property total £210 billion, £38 billion, £90 billion and £8 billion respectively as of October 2023. Some OEFs offer daily redemptions while holding less liquid assets. This means in stress, there is an incentive for investors to <u>redeem ahead of others, or for funds to struggle to meet redemption demands without fire selling assets, leading to contagion across markets.</u>	In order to be sufficiently resilient, redemption terms in OEFs should be more closely aligned with the underlying liquidity of funds' assets. In 2023, the FCA <u>published findings and wrote to asset manager CEOs</u> on liquidity management frameworks. The FSB published a <u>consultation paper</u> in July seeking to better align funds' redemption terms with the underlying liquidity of their assets. <u>IOSCO has also consulted</u> on implementing anti-dilution tools to complement this approach. FSB and IOSCO are due to finalise their recommendations by the end of 2023 for individual jurisdictions to implement.
Leverage	involves a firm increasing its exposure to a risk factor beyond what would be possible through a direct investment of its own funds.	

Vulnerability	Financial stability implications	Policy recommendations and next steps
Non-bank leverage	Leverage creates counterparty risks and can lead to sudden spikes in demand for liquidity – either to support the financing of leveraged positions, or as de-leveraging leads to forced sales, which in turn could amplify shocks and lead to market dysfunction and a potential tightening in financial conditions for households and businesses. While estimating leverage is very difficult, the notional amount of non-bank investors' OTC derivatives is estimated at almost US\$90 trillion in 2022, and global NBF financial debt is estimated at approximately US\$48 trillion, or 50% of global GDP.	More comprehensive and consistent monitoring by authorities is needed to mitigate the financial stability risks from non-bank leverage and ensure it is appropriately managed (see November 2018 FSR). The PRA has published Dear CEO letters reviewing global equity finance businesses (2021) and fixed income financing businesses (2023) including recommendations for firms to review their internal risk cultures. The PRA will follow up on firms' remediation plans via ordinary supervisory channels. The FSB has recently published a report on NBF financial leverage. In 2024, it plans work to enhance monitoring, close data gaps and to consider the wider set of policy measures to contain leverage outlined in that report.

Vulnerability	Financial stability implications	Policy recommendations and next steps
Liability-driven investment (LDI)	LDI funds' use of leverage makes them vulnerable to gilt yield shocks which, in the absence of recapitalisation, can trigger fire sale dynamics and amplify shocks, as seen during the September 2022 gilt market stress. This risked further market dysfunction and an excessive tightening of financing conditions to UK households and businesses. The total volume of UK Defined Benefit scheme liabilities hedged via LDI products is over £700 billion.	In March 2023, the FPC set out recommendations on steady-state minimum levels of resilience for LDI funds to ensure that they can absorb a severe but plausible historical stress, over the period of time needed to recapitalise the fund, without the need for forced asset sales. The FPC Recommendations have been reflected in TPR and FCA guidance to firms. The resilience framework for LDI has been functioning as intended in a higher interest rate environment, although there remain some areas for further improvement. Work is underway to put in place a monitoring and enforcement framework for pooled funds by the FCA and relevant international regulators.

Vulnerability	Financial stability implications	Policy recommendations and next steps
Margin calls	Margin can increase rapidly in stress to match the increase in expected losses and risks. This ensures that counterparty risk is properly mitigated but requires counterparties to find additional liquid assets at a time when it is more difficult for them to do so. Increases in margin that are unpredictable or unexpectedly large can cause liquidity strains on market participants and the financial system. For example, during the March 2020 ‘dash for cash’, initial margin requirements at UK central counterparties (CCPs) grew by around £58 billion, with a maximum daily increase of £10 billion; and average daily variation margin calls were five times higher than in January and February 2020.	There is a need to strengthen market participants’ preparedness to meet margin calls, as identified by a joint BCBS-CPMI-IOSCO report on margin practices. Further policy work is ongoing to improve initial and variation margining practices, to both limit the potential impacts of margin procyclicality, and better prepare market participants for jumps in margin requirements. Standard-setting bodies are developing recommendations and best practice guidance across cleared and non-cleared markets. The FSB has committed to issuing policy recommendations for consultation in 2024 on the liquidity preparedness of market participants for margin and collateral calls.